

Find It and Fix It: Counting Detective

Answer Key

Kindergarten–Grade 1

Quick Answers

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|--|---|
| 1. 11 | 5. 52, 53, 54 |
| 2. first empty box = 47, second empty box = 49 | 6. first empty box = 93, second empty box = 98, |
| 3. missing number = 20, — | number that replaces 69 = 96, — |
| 4. correct number = 40, — | |
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What is verified

3 of 6 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 3, 4, 6. The worked solution states what a correct response must contain.

Curriculum

Level: Kindergarten–Grade 1

K.CC.A.2 – *problems 1, 2, 3, 4, 6*

K.CC.C.7 – *problem 5*

Difficulty 1–4 of 5 across 12 tagged checks.

Common wrong answers

1. *If they got 10: copied the number just before the empty box instead of counting on one more*
 3. *If they got 21: wrote 21, the number Mia said next, instead of the number she skipped*
 4. *If they got 30: went back to the start of the tens after 39 instead of starting the forties*
 6. *If they got 69: kept the printed card: wrote the digits of 96 in reversed order as 69*
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1. Number track 8–13 with one empty box (between 10 and 12).

Solution: Read the number just before the empty box: 10. Counting on, one more than 10 is 11 — and 11 comes right before 12, so it fits. 11

2. Number track 45–50 with two empty boxes.

Solution: One more than 46 is

47

One more than 48 is

49

Check each answer against the number that follows it: 47 comes before 48, and 49 comes before 50. Both fit.

3. Find and fix: Mia counted 18, 19, 21, 22 and left one number out.

Solution: Whisper-count and match: “18, 19, 20, 21, 22.” Mia jumped from 19 straight to 21, so the missing number is

20

What Mia should remember (instructor-judged): every number gets said in order — after 19 comes 20, then 21. Accept any wording that names the skipped number or says “don’t skip any numbers.”

4. Find and fix: Leo counted 38, 39, 30 — one number is wrong.

Solution: The X goes on 30. After a number ending in 9 a *new* tens group starts: one more than 39 is

40

What went wrong (instructor-judged): Leo slid back to the start of the thirties instead of starting the forties. Accept any wording that says the forties begin after 39 (“after 39 comes 40, not 30”).

5. Three mixed-up cards: 54, 52, 53 — write them in counting order.

Solution: Whisper-count from 51: “51, 52, 53, 54.” The card said first is least. In counting order the cards read

52, 53, 54

6. Challenge: hundred-chart row 91–100 with two empty boxes and 69 in the wrong spot.

Solution: One more than 92 is

93

One more than 97 is

98

The box after 95 must hold one more than 95, which is

96

(69 shows the same two digits in reversed order — an easy mix-up to spot.)

Why 69 cannot be there (instructor-judged): counting on from 95 gives 96, not 69 — and 69 is much smaller than 95, so it belongs far earlier in the count. Accept either the counting-on reason or the size reason.